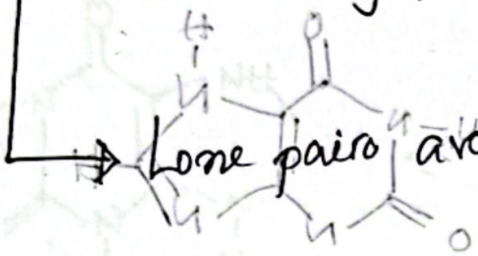
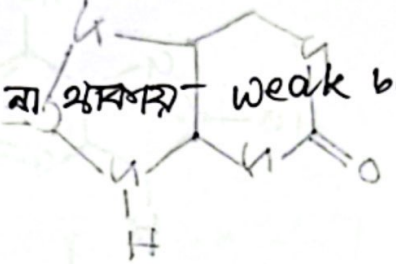
Purine & Nucleic Acid

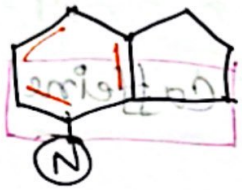
Date: 23.05.24

Purine: Organic heterocyclic aromatic compound that consists of two fused ring together.



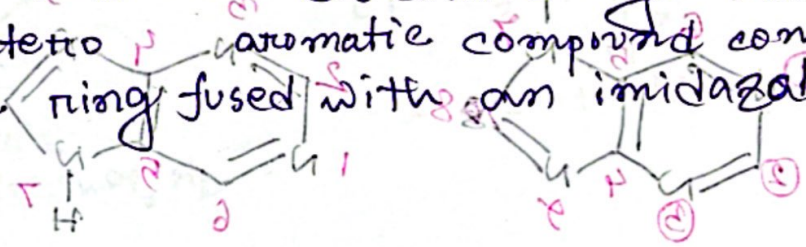
→ Lone pair available বা থাকায় weak base.

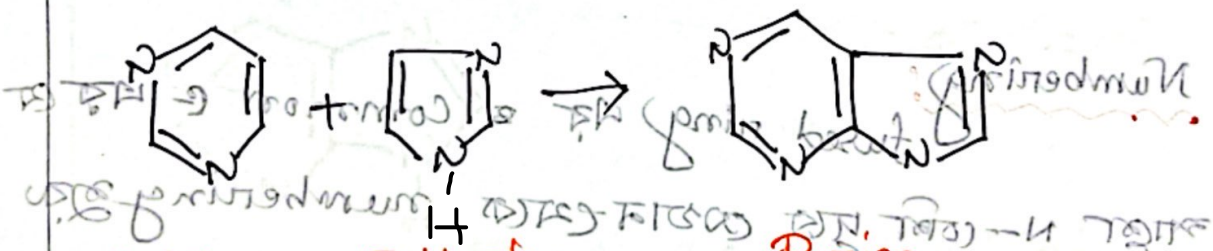




← Base (Basicity হ্রাস পায়)

Purin is the parent structure of a group of cyclic diuriedes and was discovered by Emil Fischer. It is a hetero aromatic compound consist of a pyrimidine ring fused with an imidazole ring.

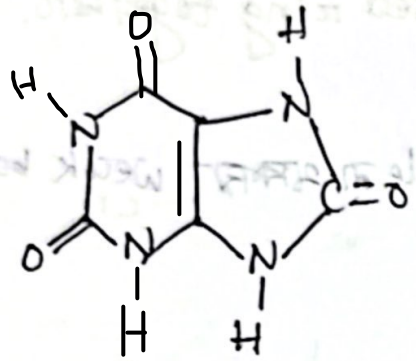




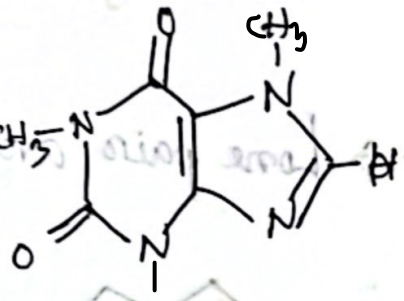
Pyrimidine + Imidazole → Purin

Nucleic Acid : DNA, RNA

Derivate of Purines:

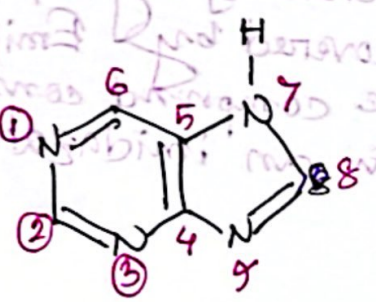


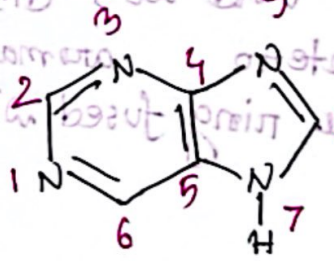
Uric acid



Caffeine

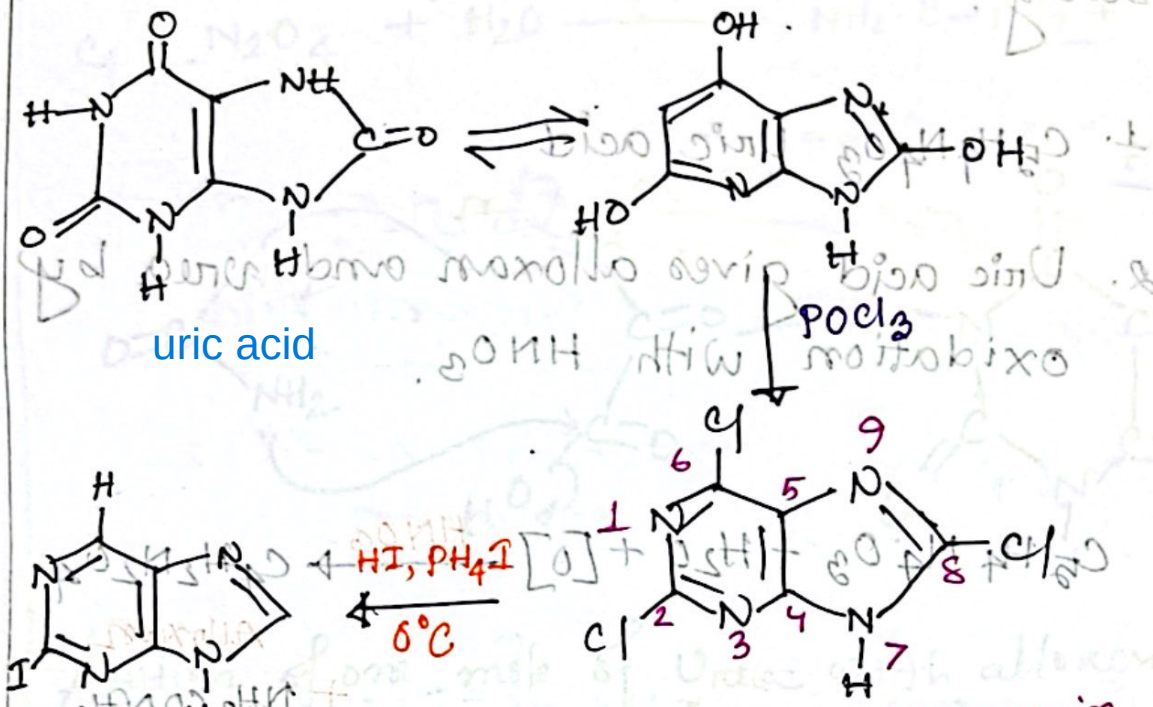
Two tautomeric form of purine

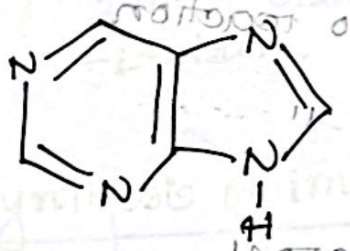




Numbering: fused ring এর ক্ষেত্রে common C ধরে যে পাশে N-টেশি দূরে সেখান থেকে numbering শুরু করে যেদিকে N আছে সেইদিকে যেতে হবে। Then, পরের ring-র যে N এর পাশে H যুক্ত সেখান থেকে শুরু করতে শ হবে।

Synthesis of Purine:





Purin.

Polymer of Nucleotides

Nucleoside + PO_4^{3-}

Base + Sugar (Deoxyribose / ribose)

(Purin / Pyrimidine)

#Find out the structure of Uric acid by degradation study:

1. $C_5H_4N_4O_3$ Uric acid

2. Uric acid gives alloxan and urea by oxidation with HNO_3.

$$C_5H_4N_4O_3 + H_2O + [O] \xrightarrow{HNO_3} C_4H_2N_2O_4 + NH_2CONH_2$$

Alloxan + Urea

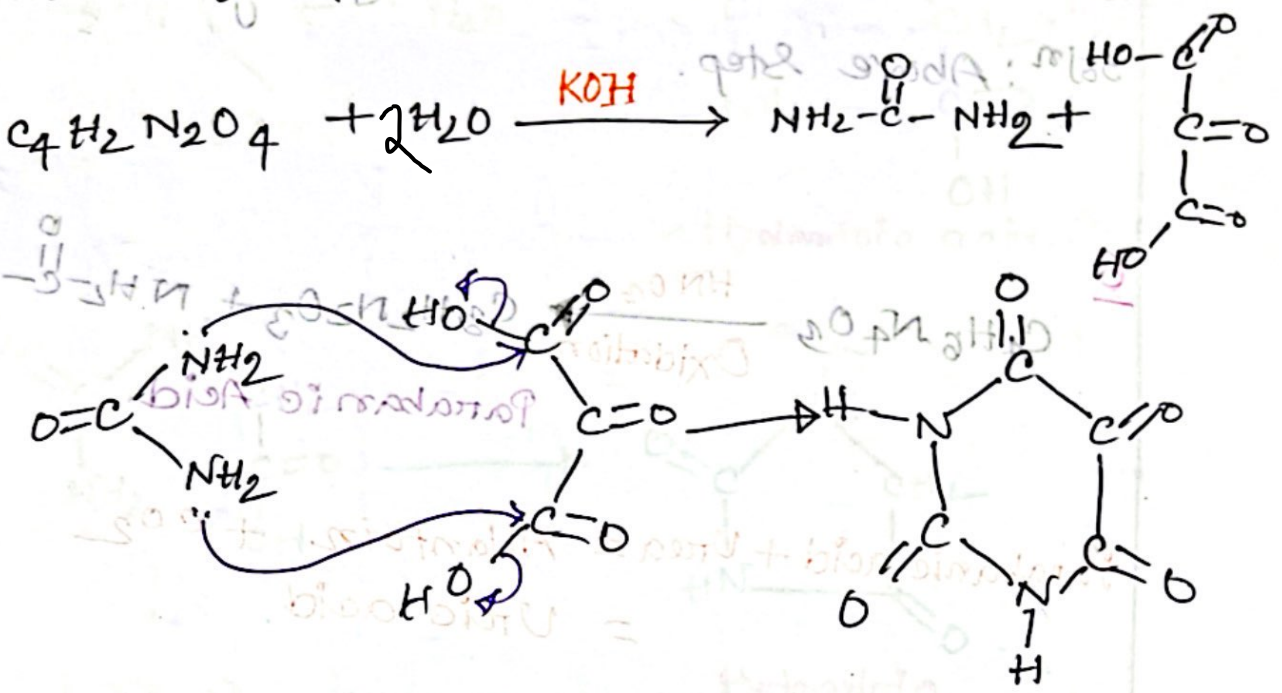
$$C_4H_2N_2O_4 + HNO_2 \longrightarrow \text{No reaction}$$

$$\text{"} + NaHCO_3 \longrightarrow \text{"}$$

Therefore, No free $-NH_2$ & $-COOH$

(3) In alloxan, there is no free amino or carboxylic acid group when alloxan is hydrolysed with alkali it produces one mole of urea and one molecule of messo

oxalic acid.



Addition of one mole of Urea with alloxane gives the structure of uric acid.

Q-1: 1 mole Urea + 1 mole meso oxalic acid = Alloxane -Explain.

Synthesis of important derivative of Purin Base:

④ $C_5H_4N_4O_3 \xrightarrow[\text{Hydrolysis}]{KMnO_4} C_4H_6N_4O_3 + CO_2$

Allantoin.

⑤ $C_4H_6N_4O_3 \xrightarrow[\text{Hydrolysis}]{KOH} 2NH_2-\overset{\overset{O}{\|}}{C}-NH_2 + \underset{\underset{COOH}{|}}{\overset{\overset{H}{|}}{C=O}}$

Glyoxylic acid.

Q: Allantoin is Divride of Glyoxalic acid — Explain

Sol^n: Above Step.

B

$$C_4H_6N_4O_3 \xrightarrow[\text{Oxidation}]{HNO_3} C_3H_2N_2O_3 + NH_2-\overset{\overset{O}{\|}}{C}-NH_2$$

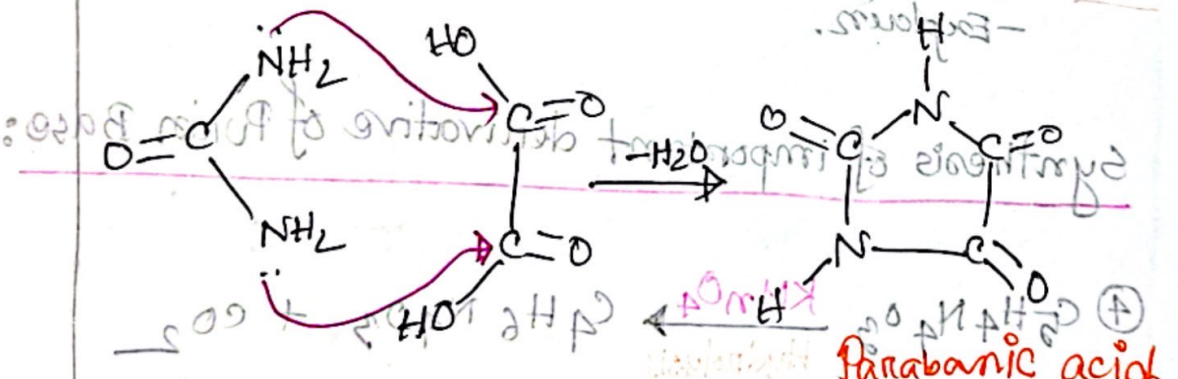
Parabanic Acid

Parabanic acid + Urea = Allantoin + CO_2

= Uric acid.

(A) $C_3H_2N_2O_3 \xrightarrow[\text{Hydrolysis}]{H_2O} NH_2-\overset{\overset{O}{\|}}{C}-NH_2 +$ COOH / COOH

Urea, oxalic acid.



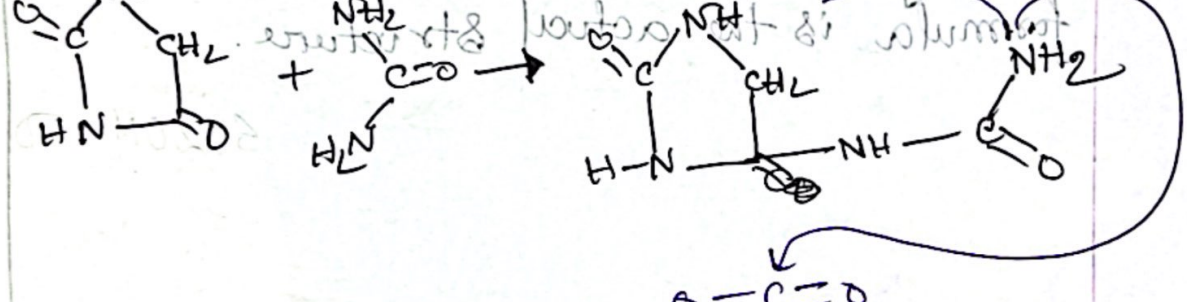
Parabanic acid.

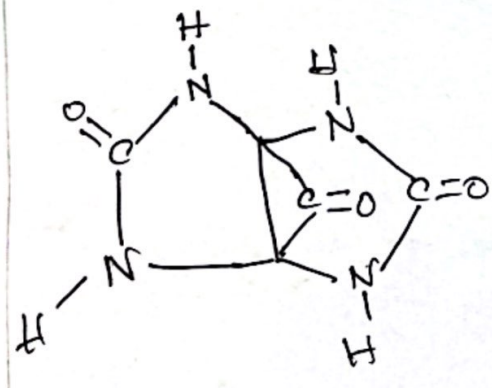
(B) $C_4H_6N_4O_3 \xrightarrow[-HI\ (100\%)]{\text{Reduction}} NH_2CONH_2 + C_3H_4N_2O_2$

Allantoin — Urea — Hydantoin.

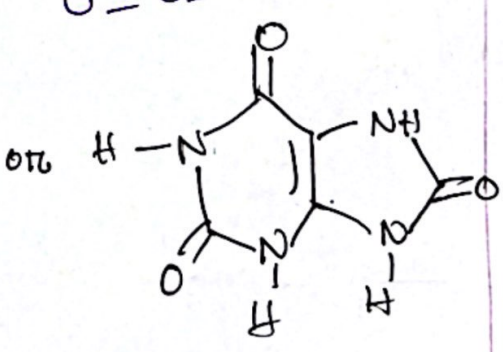
(9) $C_8H_4N_2O_2$ ⟵ (Hydrolysis, H_2O) ⟶ Hydantoic acid.

(10) Hydantoin





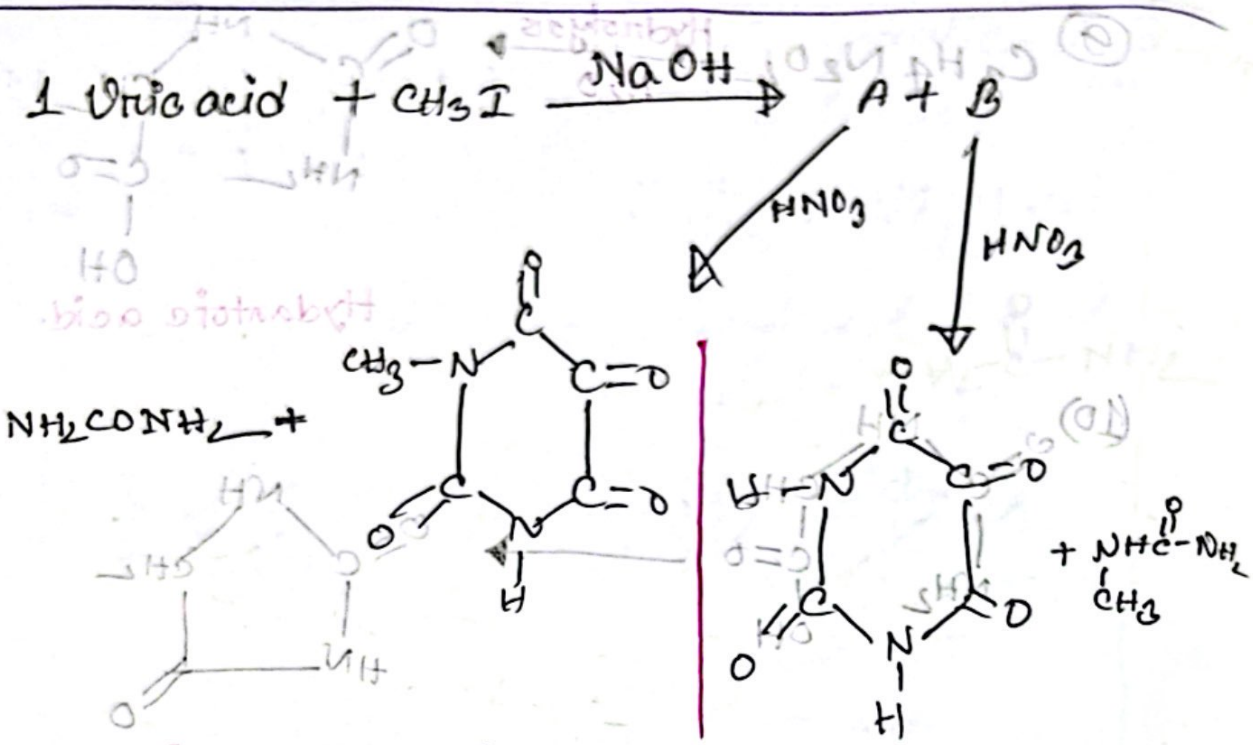
Fittig formula



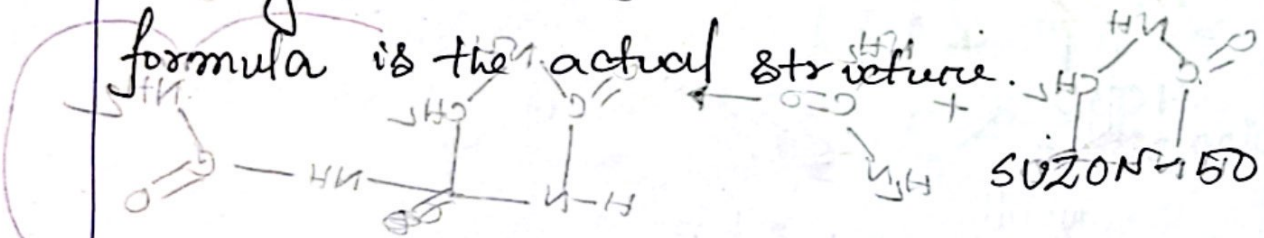
Actual structure

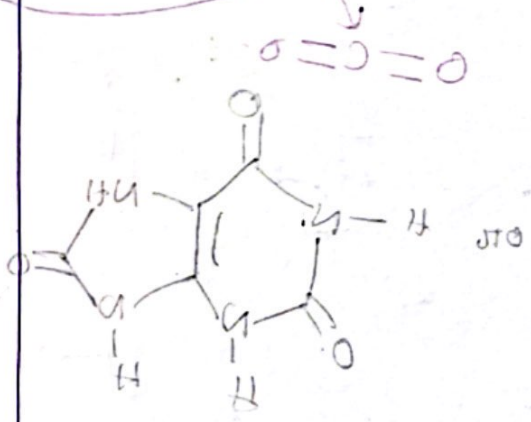
Medicens formula

1 Uric acid + CH_3I $\xrightarrow{NaOH}$ A + B

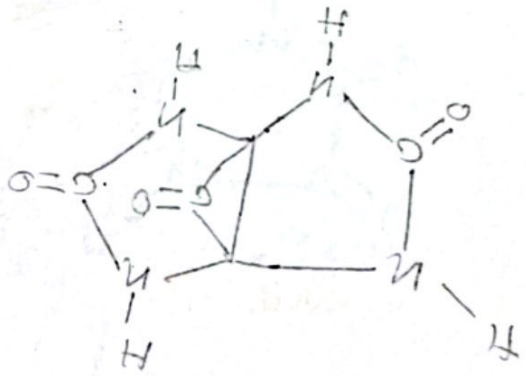


We get mixture of product. So medicus formula is the actual structure.





Actual structure

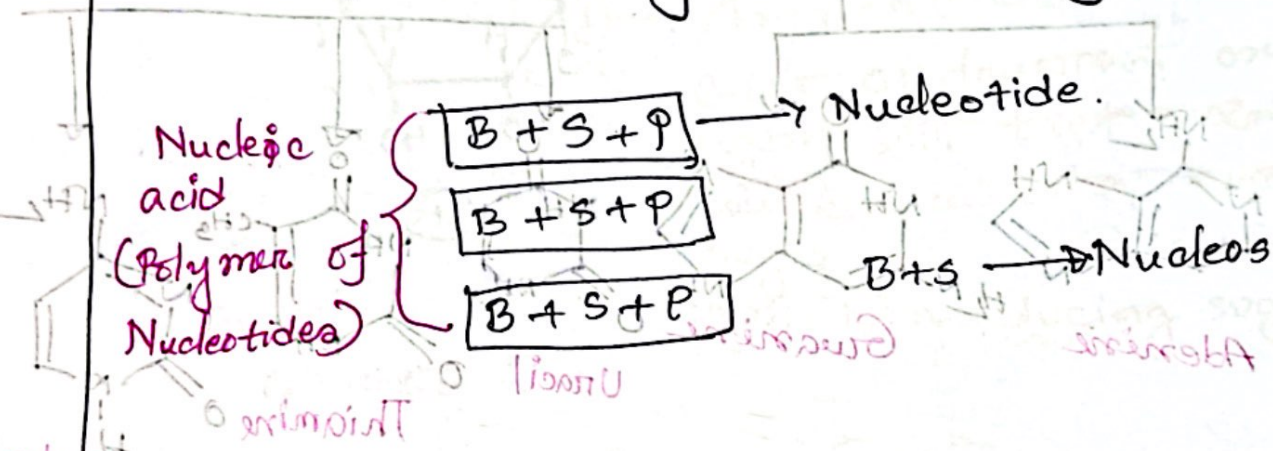


Medicus formula

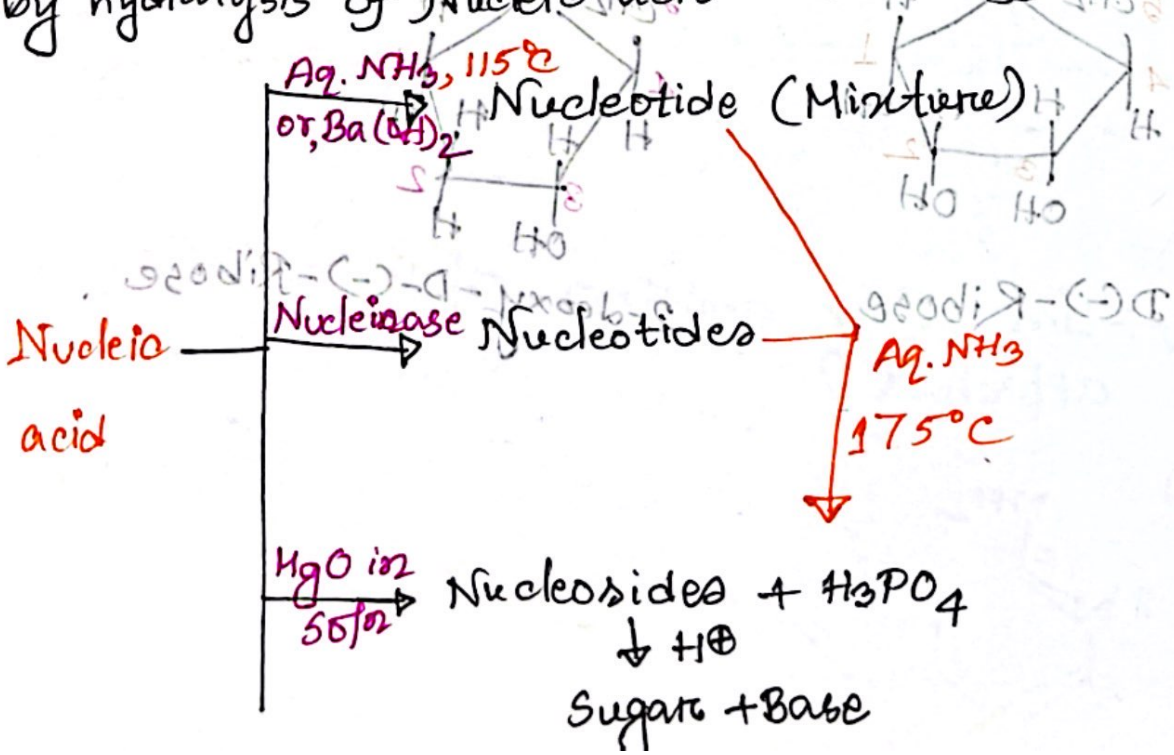
Fitting formula

Nuclei acid : Nucleic acids are high molecular weight polyme
, colourless amorphous solid. They are polymers of nucleotid
(নিরাকার) (বেশিরভাগ unit ভিন্ন shape এর)

Nucleic acid is consist of base (Purine and pyrimidine) and pentose sugar (Ribose + 2-deoxyribose) att-ached with an inorganic H_3PO_4 group.



The following chart shows the nature of products obtained by hydrolysis of Nucleic acid under different conditions.

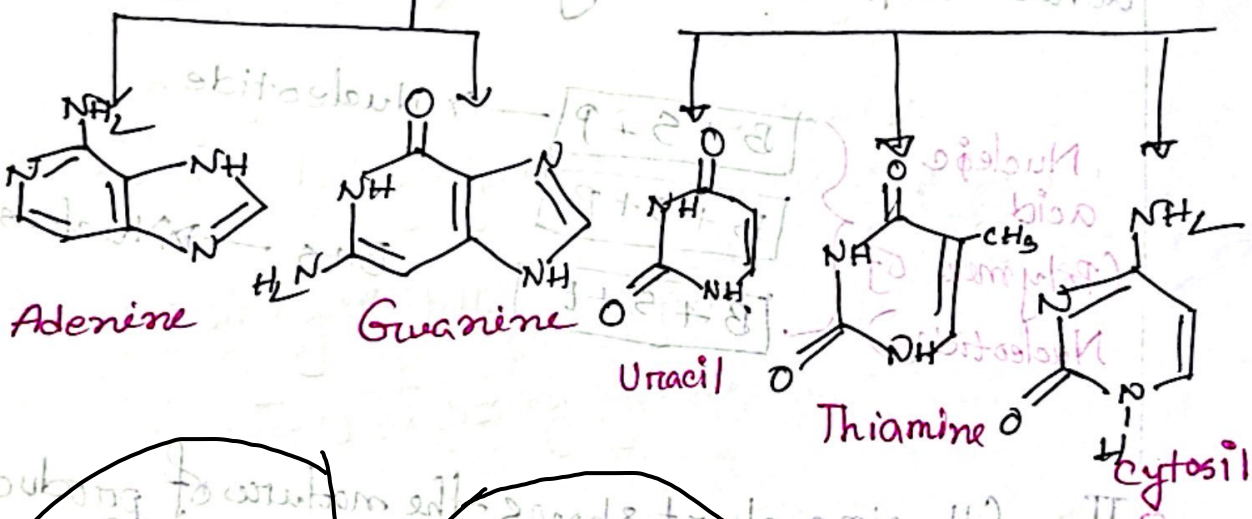


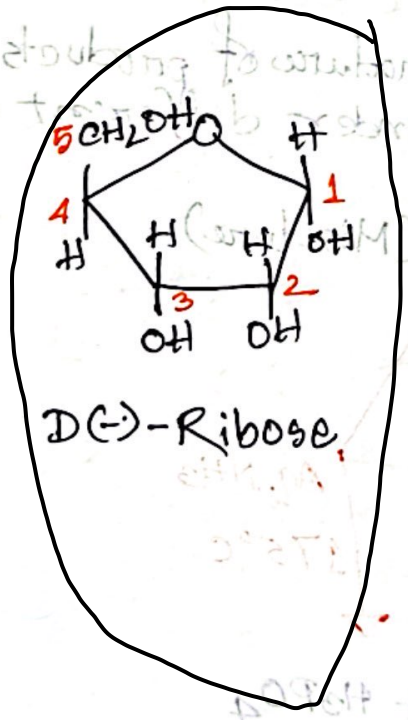
Nucleic acid ⟶ Polymer of Nucleotide

Nucleotides ⟶ Nucleosides + H_3PO_4.

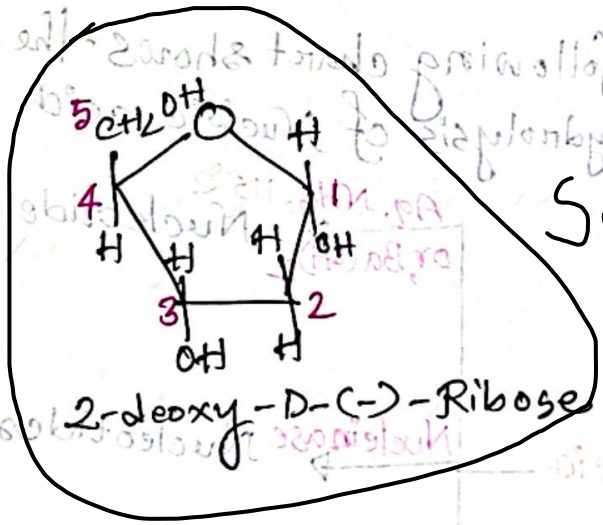
Nucleoside ⟶ Pentose sugar + Base (~~Ribose/2-Deoxyribo~~

Purin + Pyrimidine.





D(-)-Ribose.



2-deoxy-D-(-)-Ribose.

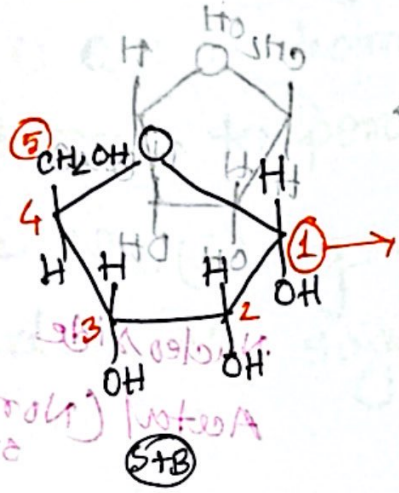
Sugars

A,T,G,C=DNA
U,A,G,C=RNA

Adenosine (Ribose + adenine)

Guanosine (Guanine + Ribose)

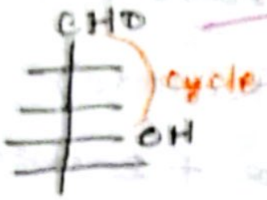
Thyamidine (Thyamine + 2-deoxyribose)



Anomaric Carbon

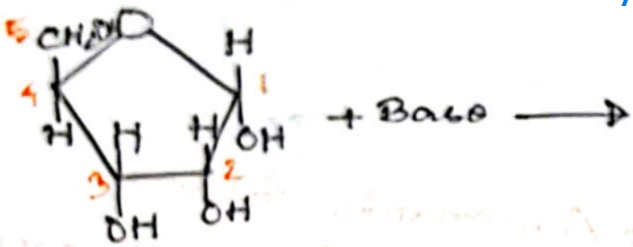
C-এর OH free থাকলে oxidizing agent দ্বারা treat করলে সহজে oxidized হয়ে reducing sugar হয়। যদি OH free না থাকে তখন Non-reducing sugar হয়।

SUZON
CHEM-50

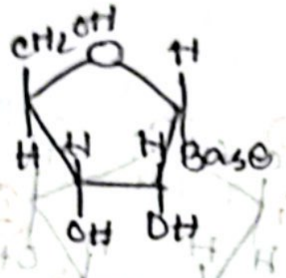


→ Reducing sugar.

Cu2+ + 5OH- fehling reagent
Ag(NH3)2)+ tolleng reagent



Ribose sugar
Hemiacetal (Reducing sugar)



Nucleoside
Acetal (Non- reducing sugar)

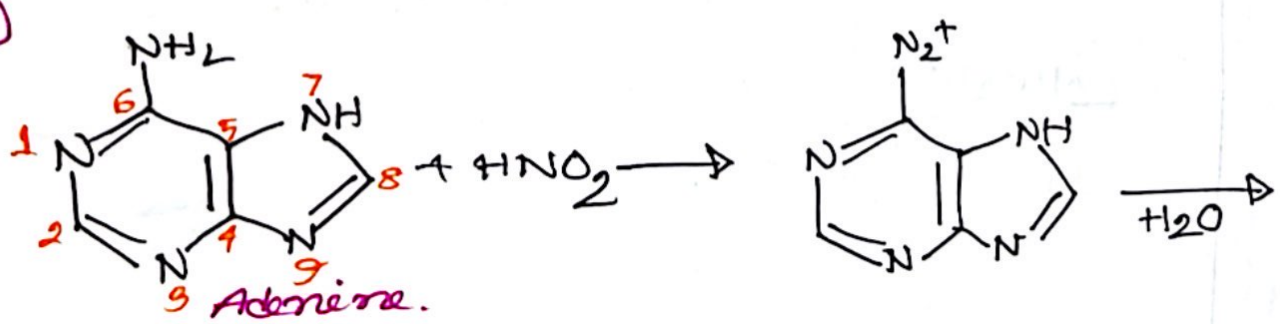
Fehling & Tollens reagent
Mild oxidizing agent.

Q→ Sugar এর কয় নম্বরে Base যুক্ত আছে? কীভাবে explain করবে?

উঃ: 1-No C. Explanation→above

Q→ In which position of Base the sugar is attached?

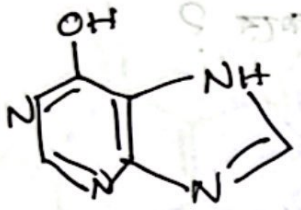
①



Adenine.

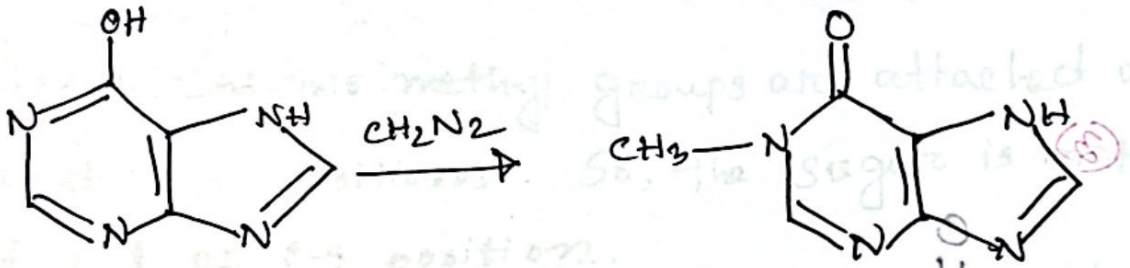
let consider a nucleoside

Scanned with CamScanner



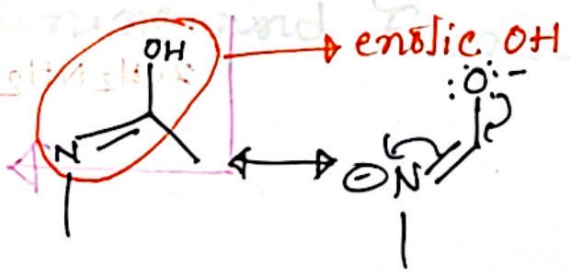
When adenine is treated with HNO_2 the NH_2(6) group is 6-C is deaminetated and evolve N_2 gas at room temperature in presence of H_2O and replace NH_2(6) by -OH group. So, 1° NH_2 is not attached to the sugar (C-1).

(2)



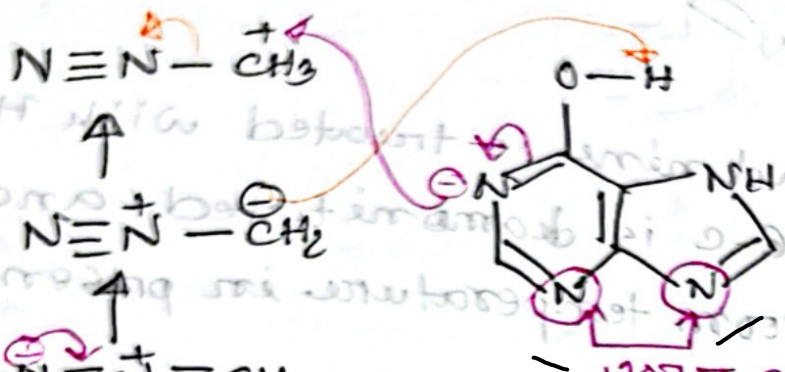
When uracil derivative is treated with diazomethane (CH_2N_2) the $-CH_3$ group attack at C-1

No sugar at C-1 of base.



Q: Diazomethane কোন N attack করে ?

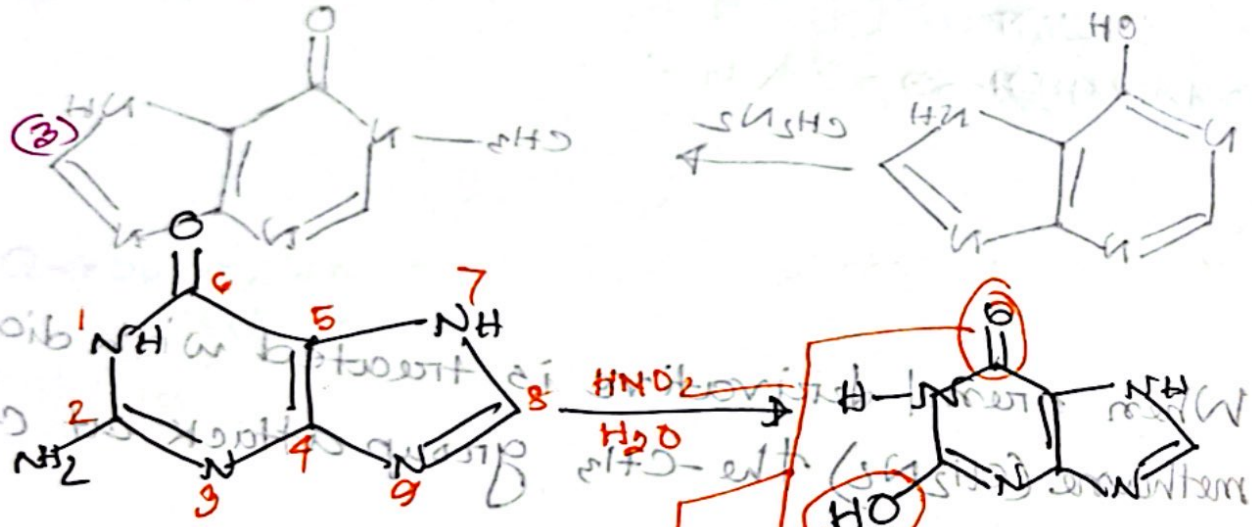
Soln:



Mechanism

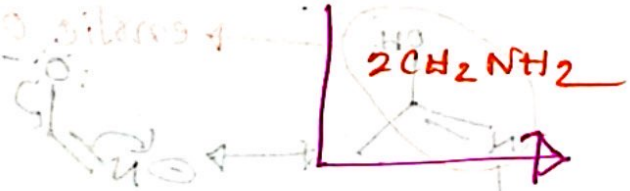
যেখানে possible না, then N-1 bond

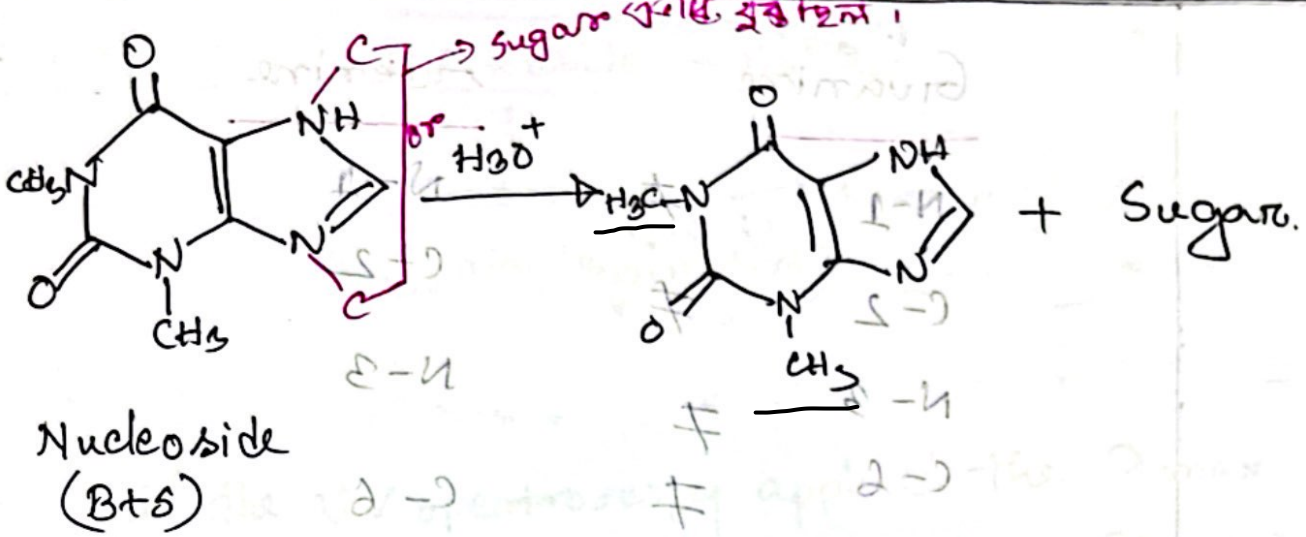
Therefore, 1- N attack করে।



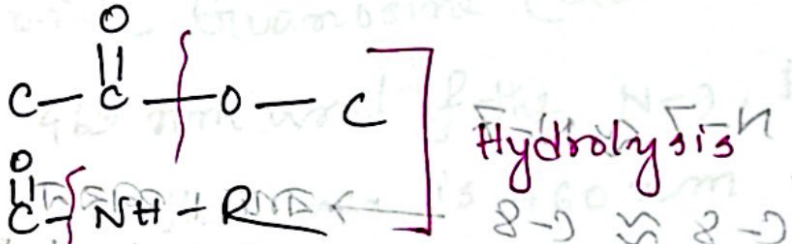
Guanosine → Nucleoside (S+B)

এগুলোর Presence এর জন্য 2nd CH_2NH_2 লাগে,





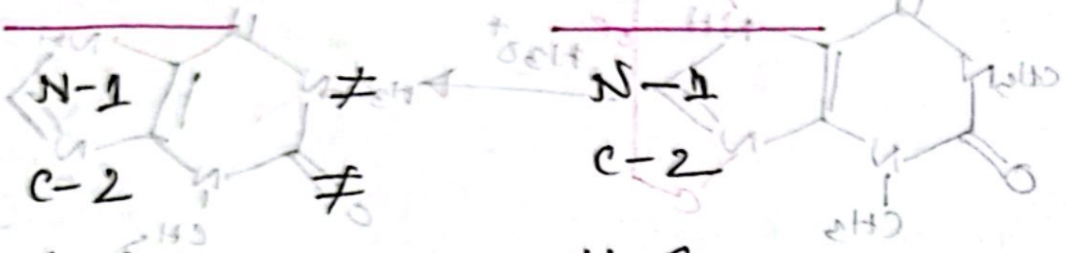
Nucleoside
(B+S)



Though the two methyl groups are attached at C-1 and C-3 positions. So, the sugar is not attached at C-1 or C-3 position.

④ Enzyme reacts in a specific site. The sugar is attached in the same position of adenine and Guanine. The equivalent and nonequivalent of adenine and Guanine are —

Guanine		Adenine
N-1	≠	N-1
C-2	≠	C-2
N-3	≠	N-3
C-6	≠	C-6



N-7 ≈ N-7

C-8 ≈ C-8 ⟶ বাদে। কারণ C-C bond কে Hydrolysis এর মাধ্যমে ভাঙা যায় না,

N-9 ≈ N-9

C-N / C-O ⟶ পেপটাইডে,

Nucleotide ⟶ Nucleoside + H_3PO_4

↓

Base + Sugar ⟶ Ribose
(C-1)
(Purin/Pyrimidine)

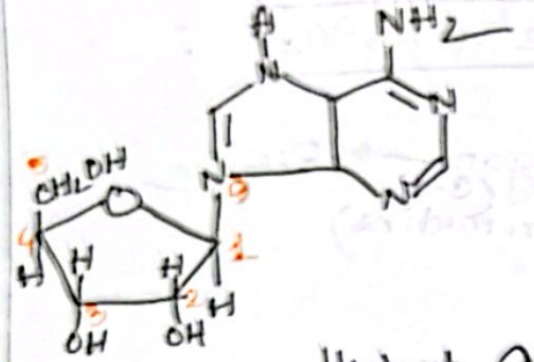


If the UV spectroscopy applied the λmax for nucleoside Guanosine (Guanine + Ribose) is found 463 nm and if the N-9, H is replaced by $-CH_3$ then the λmax is 460 nm which is almost same as the guanosine.

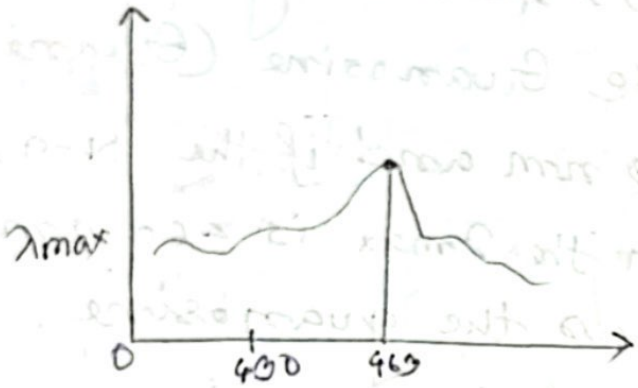
But if the N-7, H is replaced by $-CH_3$ we get,

λmax at 440 nm which is different from the guanosine.

So, the sugar (C-1) is attached at N-9 of purin base.



Highest λmax 463 nm absorb করে।



Chromophorosity → Conjugate double bond অথবা system বানিয়ে থাকে।

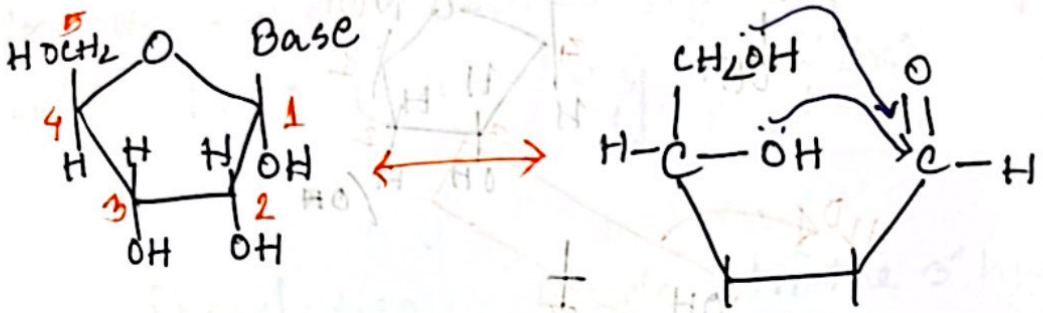
In which part of Nucleoside the PO_4 is attached and How?

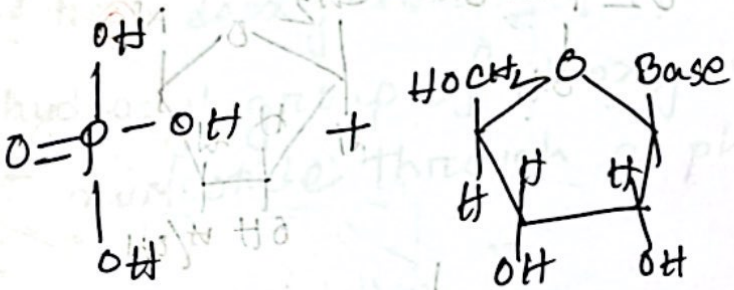
When nucleotide is hydrolysed under controlled environment we get ribose monophosphate and base.

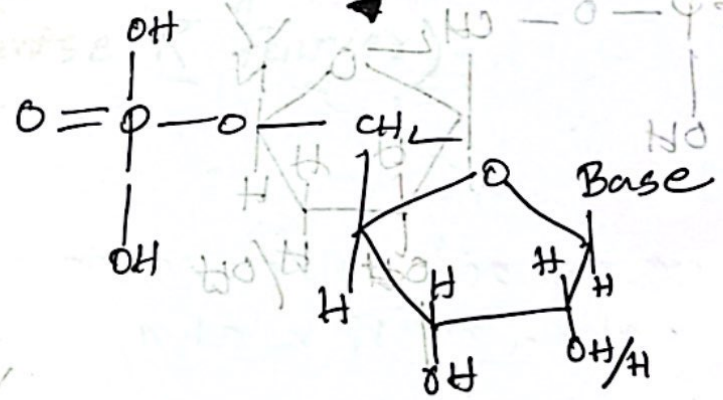
Nucleotides $\xrightarrow[\text{Hydrolysis}]{\text{Control}}$ Ribose monophosphate + Base

(এর মানে PO_4 এর সাথে যুক্ত থাকে।)

So the $-PO_4$ group is attached with the sugar part. The sugar contains $-OH$ group at 2, 3, 5 positions. So the addition of H_3PO_4 will attack at 2, 3, 5 position of sugar to add PO_4 group.



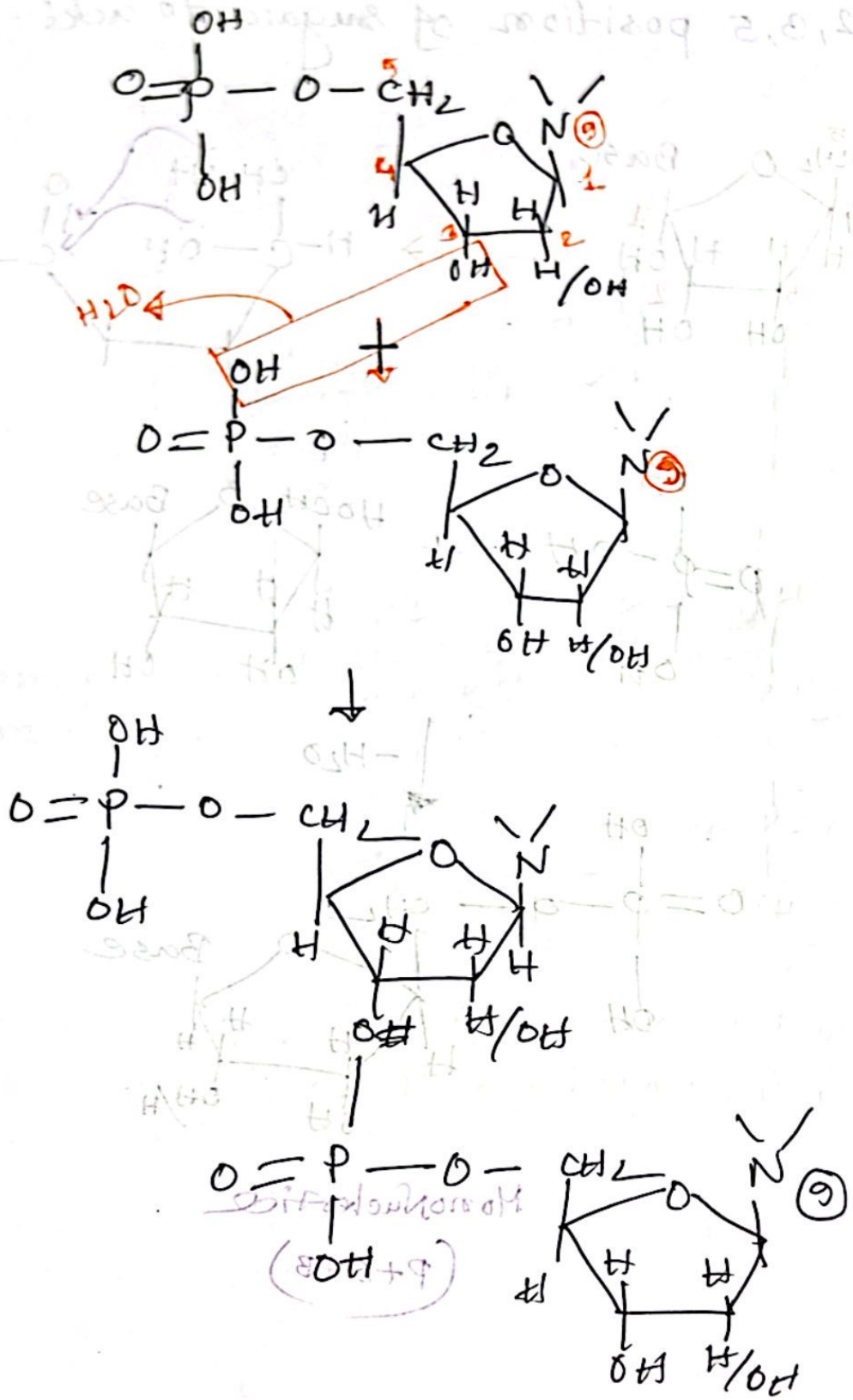




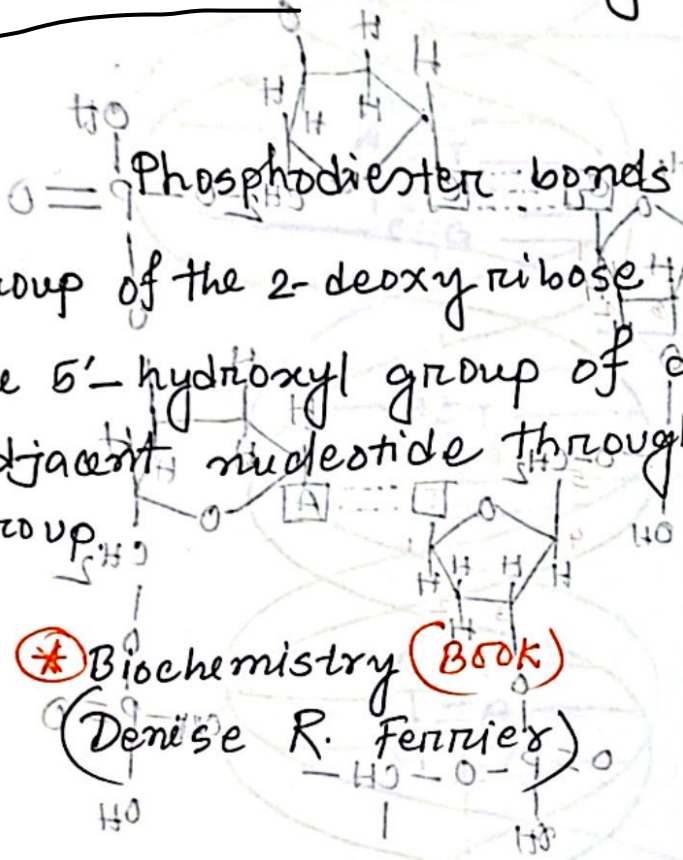
MonoNucleotide

(P+S+B)

পরপর অনেকগুলো Mononucleotide যুক্ত হয়ে Polynucleotide হয়।



DNA is a polymer of deoxyribonucleic monophosphate (dNMP) covalently linked by 3′ → 5′ phosphodiester bonds. DNA exist as double stranded molecule (dsDNA) in which two strands wind around each other forming double helix (secondary structure)

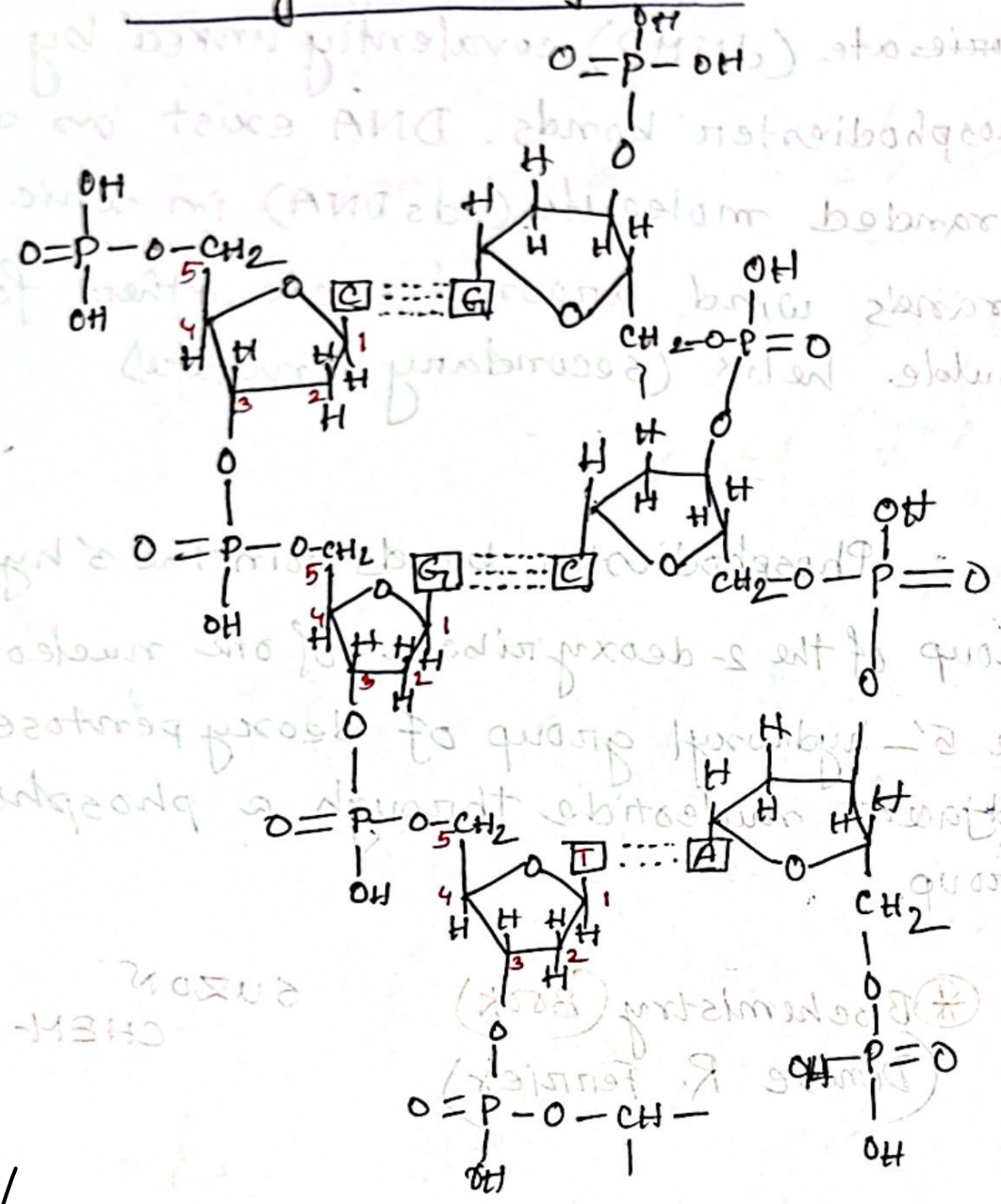


Phosphodiester bonds join the 3′ hydroxyl group of the 2-deoxyribose of one nucleotide to the 5′-hydroxyl group of deoxypentose of an adjacent nucleotide through a phosphoryl group.

(*) Biochemistry (Book)
(Denise R. Ferrier)

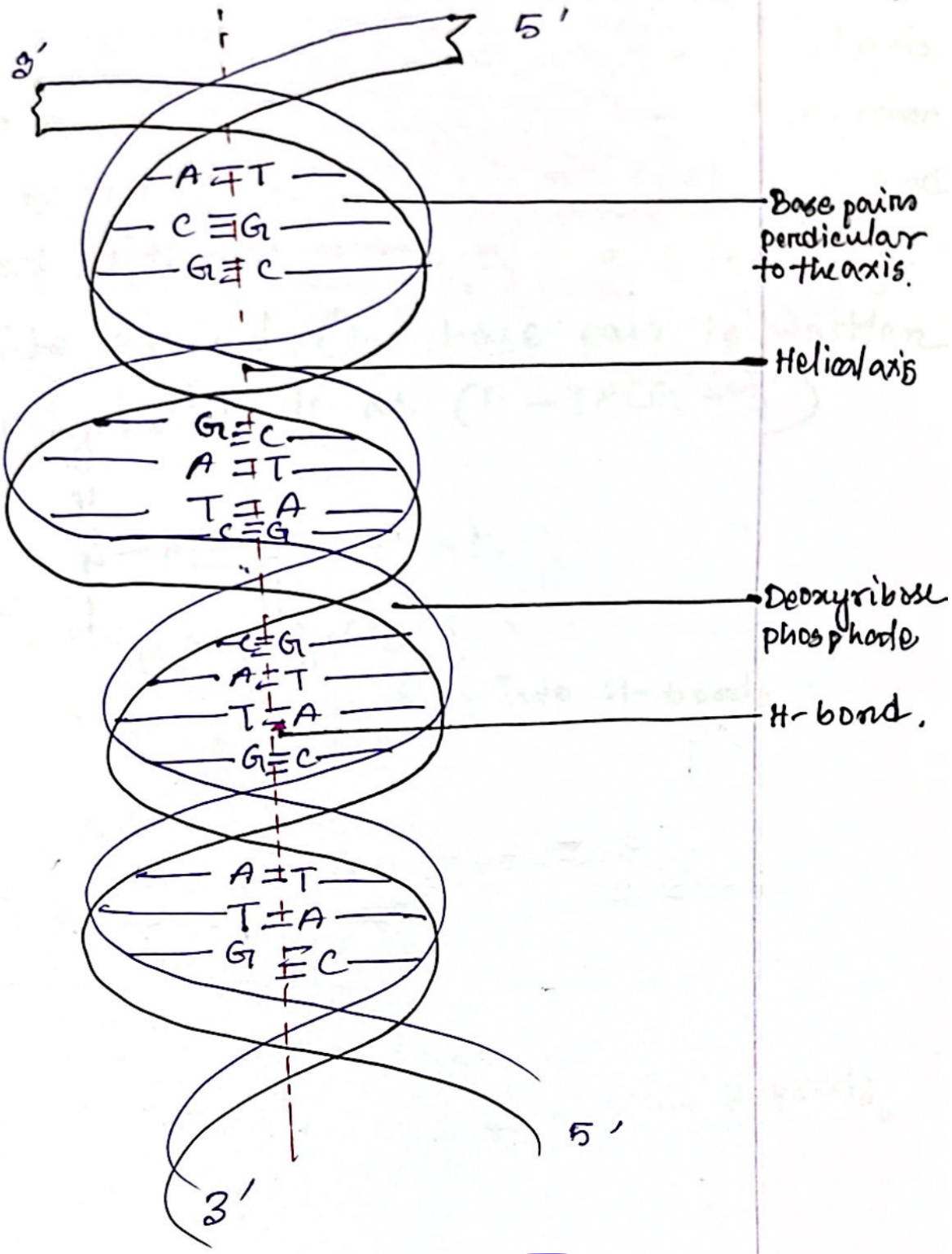
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Primary Structure of DNA



H-bond হলে হলে 20Å distance থাকে হলে। Purin-Purin distance বেশি হয় তাই H-bonding হয় না।

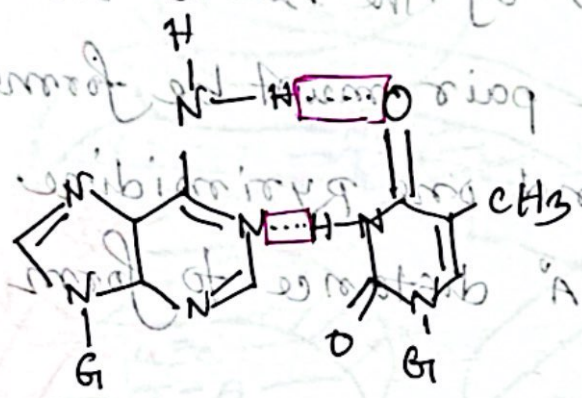
Double Helix of DNA (Secondary structure)



primary structure means full structure with base +sugar+phosphate

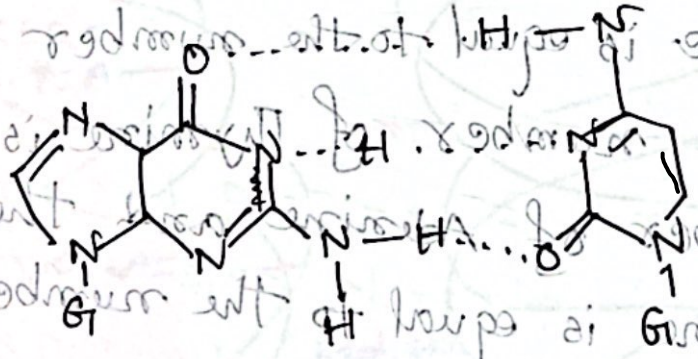
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In the double helix, the two chains are coiled around a common axis called the helical axis. The chains are paired in an antiparallel manner i.e the 5' end of one polynucleotide strand is paired with the 3' end of the other polynucleotide strand. One base pair is written in a polynucleotide as (5'-TACG-3')



Two H-bonds.

Adenine Thymine



Three H-bonds.

Guanine Cytocine

Complementary base pairs in nucleic acid.

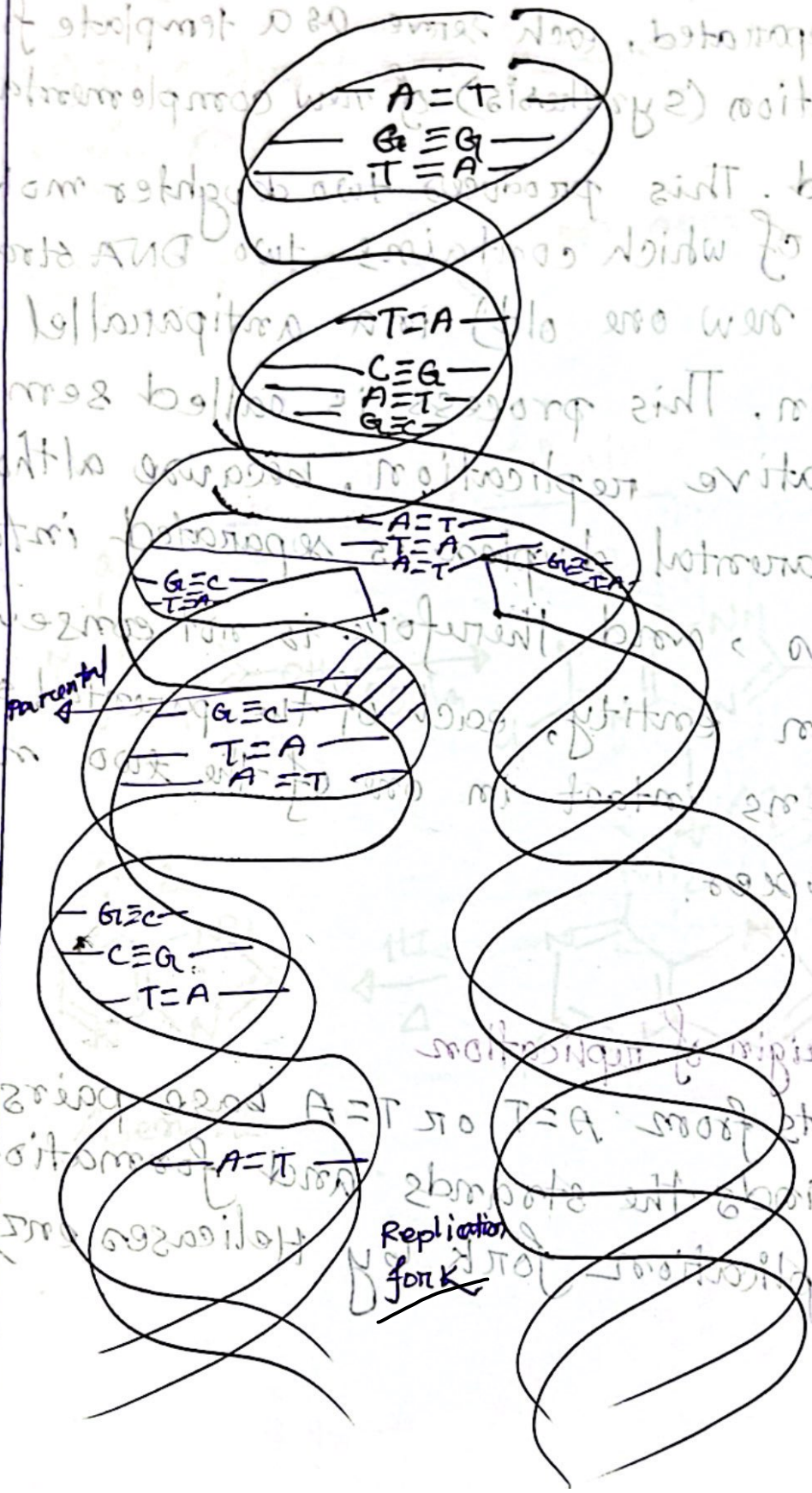
In the DNA double helix the hydrophilic deoxyribose – phosphate backbone of each chain is on the outside of the molecule whereas the hydrophobic bases are stacked inside. The overall structure resembles a twisted ladder. The bases of one strand of DNA are paired with the bases of the second strand by H-bonding. The pair must be formed between one purin and one pyrimidine base to maintain 20 A° distance to form H-bonds.

V.V.I.

Chargaff rule:

In a nucleotide the number of purin base is equal to the number of pyrimidine base. The number of Thymine is equal to the number of Adenine and the number of Guanine is equal to the number of cytosine. This is called chargaff rule.

Replication of DNA



Replication of DNA:

When the two strands of ds DNA (double strand DNA) are separated, each serve as a template for the replication (synthesis) of new complementary strand. This produces two daughter molecules, each of which contains two DNA strand (one new one old) in a antiparallel orientation. This process is called semiconservative replication, because although the parental duplex is separated into two halves, and therefore is not conserved as an entity, each of the parental strands remains intact in one of the two new duplexes.

(i) Origin of replication

starts from A=T or T=A base pairs, unwinds the strands and formation of replication fork by Helicases enzyme.

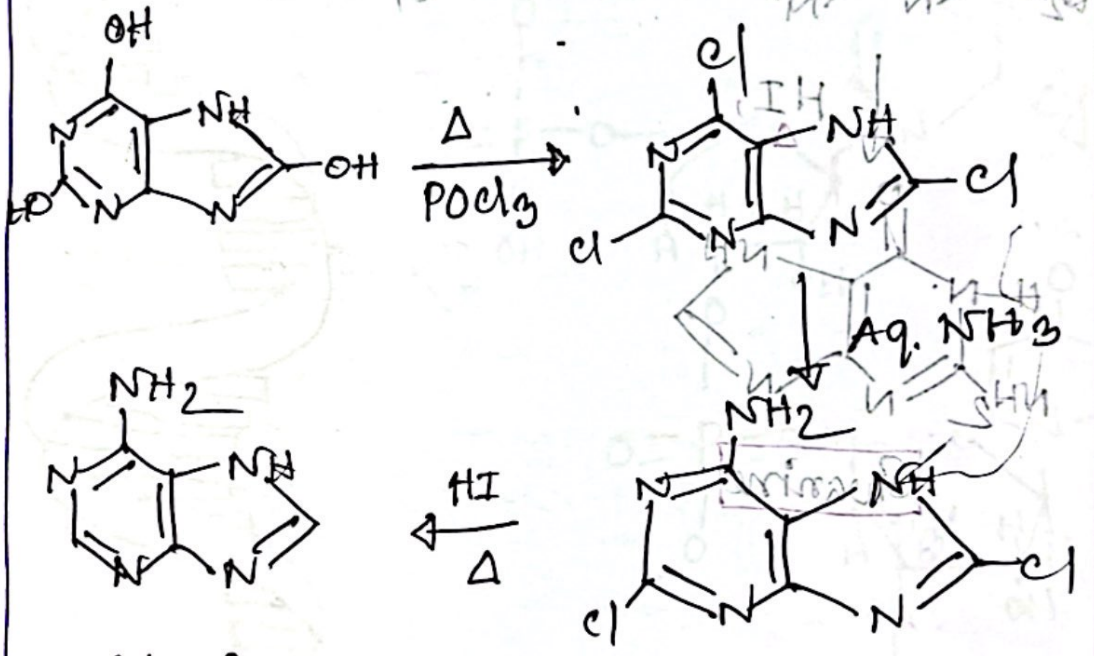
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(ii) HDP (Helex destability proteins) dosn't let the single strand to form double helix.

(III) DNA direct RNA primers synthesis by RNA polymerase.

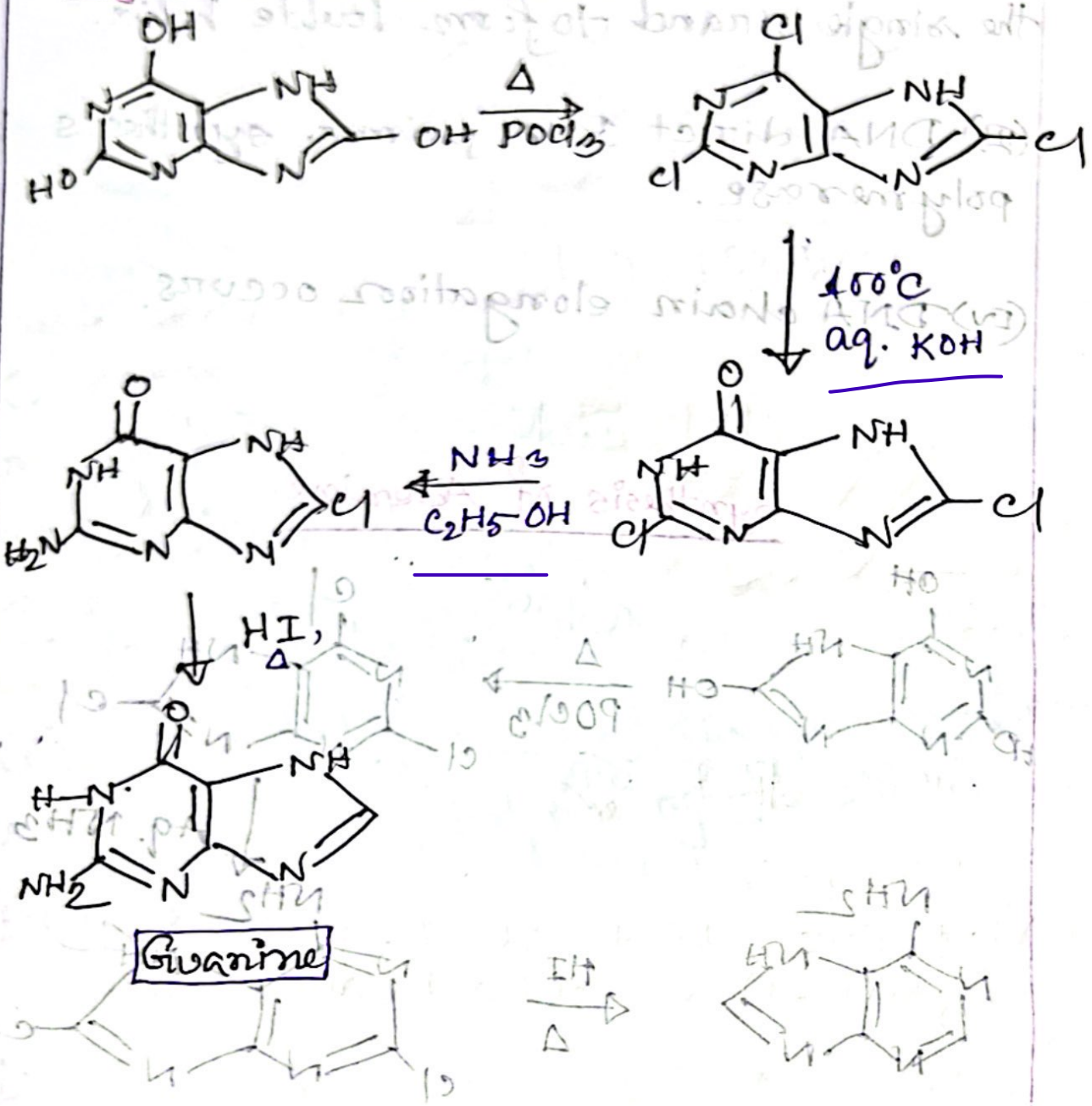
(IV) DNA chain elongation occurs.

Synthesis of Adenine

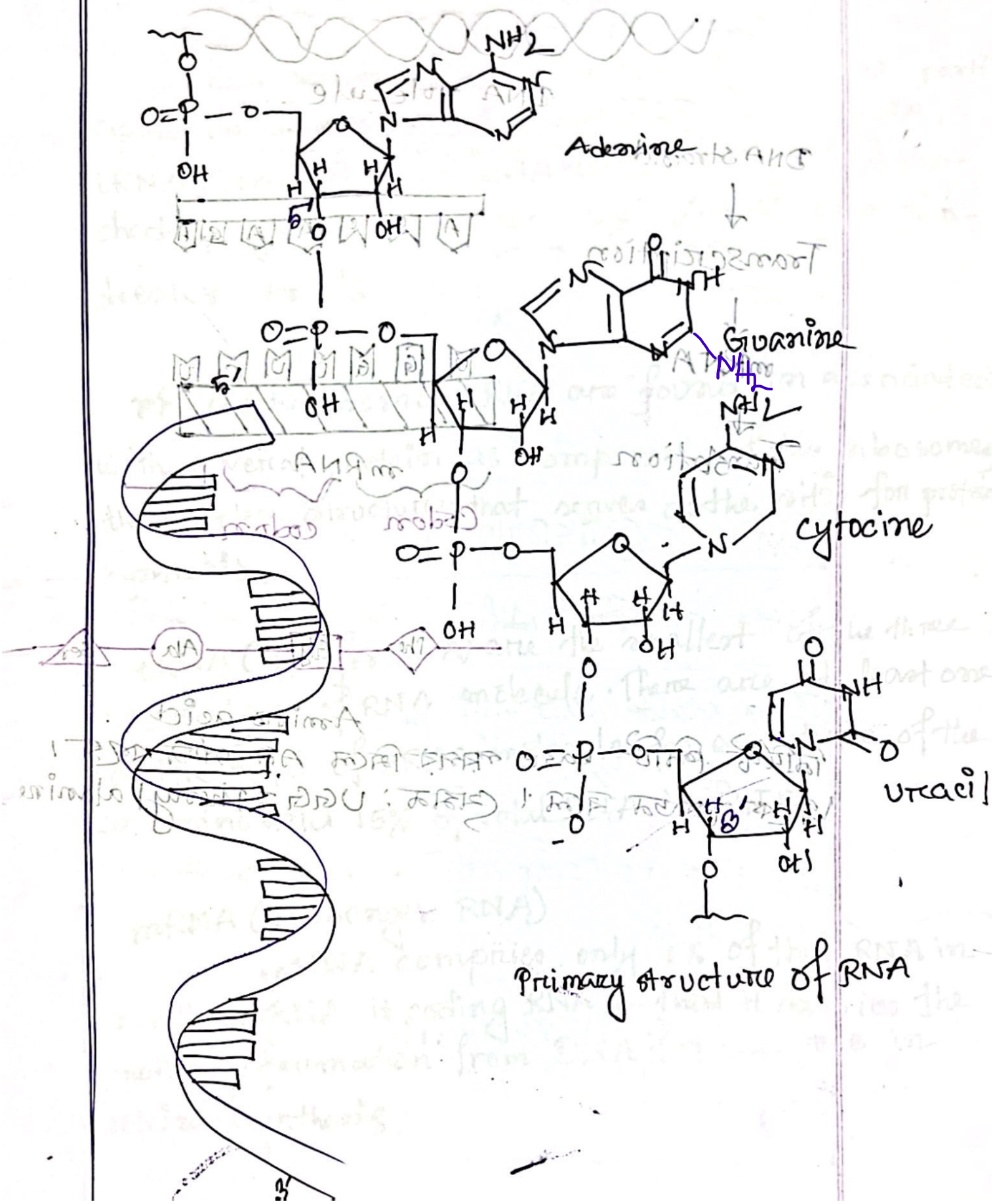


Adenine

Synthesis of Guanine



RNA (Ribonucleic acid)



Primary structure of RNA



DNA Molecule.

DNA Strand

↓

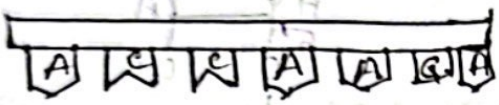
Transcription

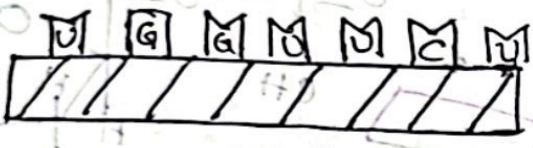
↓

mRNA

↓

Translation





mRNA

Codon codon.



Amino acid

নির্দিষ্ট তিনটি base পরপর মিলে AA গঠন করে।
একে কোডন বলে। যেমন: UGG = Phenylalanine.

Types of RNA

There are three major types of RNA, that participate in the process of protein synthesis, rRNA, tRNA, mRNA. Like DNA these RNA are unbranched polymeric together by 3'—5' phosphodiester bonds.

rRNA (Ribosomal RNA are found in associated with several protein as components of the ribosomes the complex structures that serves as the sites for protein synthesis.

tRNA (Transfer RNA) are the smallest of the three major types of RNA molecule. There are at least one specific type of tRNA molecule for each type of the 20 amino acid 15% of total RNA is tRNA.

mRNA (Messenger RNA)

mRNA comprises only 5% of that RNA in the cell. mRNA is coding RNA in that it carries the genetic information from DNA for the use in protein synthesis.

Difference Between RNA VS DNA

DNA	RNA
Sugar moiety is Deoxyribose	Sugar moiety is Ribose
The bases present are Adenine, Guanine, cytocine & Thymine	The bases are Adenine, Guanine, cytocine and uracil.
Double stranded molecule	Single stranded molecules
Obeys chargaff's rule	Does not obey chargaff's rule
Bases are not modified	Bases are modified
It is stable and not hydrolyzed by alkalis	It is unstable and hydrolyzed easily by alkalis.
Life time is comparatively high.	RNA is short lived.
Present in the nucleus, mitochondria and chloroplast	Present in the nucleus, mitochondria, nucleolus, ribosomes and cytosol.

Additional RNA, these RNA performs specialized structural, catalytic and regularly functions and are not translated are called non-coding (RNA).

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